Drop2Crop — System User Manual

Smart Cloud IoT Precision Irrigation & Fertigation Station

Target Device: ESP32-S3 (16MB Flash, 8MB PSRAM)	**Firmware Version:** v2.4-Production
Cloud Protocol: MQTT over Secure WebSockets (WSS)	**Broker:** test.mosquitto.org (Port 8081 / 1883)
Dashboard URL: https://fahimashabb.github.io/drop2crop/	**Timezone:** Asia/Dhaka (BST, UTC+6)

1. System Overview & Architecture

Drop2Crop is an automated precision agricultural controller engineered to eliminate manual irrigation guesswork, prevent overwatering and root asphyxiation, conserve reservoir water, and protect crop yields across all seasons. The system pairs an industrial-grade ESP32-S3 microcontroller running real-time sensing firmware with a zero-cost, high-speed cloud dashboard hosted on GitHub Pages. You can monitor and control your farm from anywhere in Bangladesh (e.g., from Chittagong, Dhaka, or abroad) while the device operates unattended in the field (e.g., in Rajshahi).

2. Zero-Code WiFi Setup & Relocation (Captive Portal)

Drop2Crop features a fail-safe, zero-code **Captive Portal Provisioning System**. If you move your hardware to a new farm, change your router, or switch your mobile hotspot, **you never need to plug the ESP32 into a computer or re-flash code**. The device reconfigures wirelessly in less than 60 seconds.

- **Automatic Activation:** Upon boot, the ESP32 attempts to connect to its saved WiFi. If the network is unavailable for 12 seconds, it automatically creates a secure open Access Point named **Drop2Crop_WiFi_Setup**.
- **Manual Force Activation:** At any time while the device is powered on, press and hold the physical **BOOT button (GPIO 0)** on the ESP32 for 3 seconds. The onboard status LED will blink rapidly, wiping previous network locks and launching the setup portal.
- **Connect with Phone or PC:** Open WiFi settings on your smartphone or computer, find the network **Drop2Crop_WiFi_Setup**, and tap connect.
- **Portal Webpage:** The portal login page opens automatically. If it does not, open your web browser and navigate to **http://192.168.4.1**.
- **Scan & Save:** Tap your local 2.4 GHz WiFi network name, enter your WiFi password, and click **Save & Connect**.
- **Instant Cloud Reconnection:** The ESP32 saves the credentials to its onboard Non-Volatile Flash Memory (NVS), restarts its radio, connects to the Internet, synchronizes live Bangladesh Standard Time (UTC+6) via NTP, and connects to the cloud MQTT broker.

CRITICAL NETWORK NOTE: ESP32-S3 radios operate exclusively on **2.4 GHz Wi-Fi** (802.11 b/g/n). If your router broadcasts combined 2.4 GHz and 5 GHz networks under separate SSIDs, always connect Drop2Crop to the 2.4 GHz network (e.g. *'Dracarys'*, not *'Dracarys+'*).

3. Operational Modes & Intelligent Logic

Drop2Crop provides four distinct operational modes selectable on-the-fly from the cloud dashboard:

Mode	Mode Name	Operational Behavior & Safeguards
Mode 0	**Smart Auto (Soil + Rain)**	Full autonomous closed-loop irrigation. Turns pump ON when soil moisture drops below **35%**, and turns pump OFF once saturation reaches **70%**. System is automatically **inhibited for 24 hours** if rain is detected, and blocked if reservoir level is below **15%**.
Mode 1	**Alert-Only Monitor**	Passive sensing & notification mode. When soil moisture drops below 35%, the system triggers cloud dashboard alerts, audio chimes, and status warnings without activating the pump. The farmer inspects the crop and turns the motor ON manually via cloud button.
Mode 2	**Multi-Timer Schedule**	Clock-driven precision irrigation. Runs motor exclusively according to user-configured daily timer slots (e.g. 6:00 AM for 20 mins, 6:00 PM for 40 mins). Rain lockout still protects against overwatering.
Mode 3	**Manual Cloud Override**	Direct farmer control. Start or stop the pump on demand from the dashboard. Hardware dry-run reservoir protection remains active at all times to prevent motor burnout.

4. Multi-Timer Scheduling Engine

The ESP32-S3 firmware contains an autonomous hardware scheduling engine that tracks Bangladesh Standard Time (UTC+6) via Network Time Protocol (NTP). Up to **10 independent daily timer slots** can be saved into Non-Volatile Flash Storage (NVS).

- **Configurable Parameters:** Set execution hour (00-23), minute (00-59), and run duration (1 to 240+ minutes).
- **Multiple Daily Runs:** Add morning (e.g., 06:00 for 20m), afternoon (e.g., 14:00 for 15m), and evening (e.g., 18:00 for 40m) schedules simultaneously.
- **NVS Flash Persistence:** Schedules remain permanently saved even through power outages, brownouts, or hard resets.
- **Offline Autonomous Operation:** Even if the farm internet connection drops temporarily, the onboard real-time clock keeps running and executes all scheduled watering cycles on schedule.
- **Anti-Duplicate Lock:** A digital `ranToday` latch ensures a schedule only executes once per calendar day and resets automatically at midnight (00:00:00).

5. 24-Hour Rain Lockout & Reservoir Dry-Run Protection

24-Hour Rain Lockout System: When the optical raindrop sensor detects precipitation, a high-precision hardware countdown timer of exactly **86,400 seconds (24 hours)** is initiated. During this 24-hour cycle, both automatic moisture irrigation and timer runs are strictly locked out to protect plant roots from asphyxiation. The live countdown is displayed in real time on the web dashboard. A manual cloud override button is available if immediate watering is necessary.

Ultrasonic Reservoir Level Monitoring: An HC-SR04 ultrasonic sensor mounted above the water reservoir continuously measures distance to the water surface and converts it into a calibrated percentage (0% to 100%).

- **Level < 20%:** Triggers amber visual and badge warnings on the cloud dashboard alerting the farmer to refill.
- **Level < 15%:** Hard safety emergency cutoff immediately shuts off the relay to prevent pump cavitation and impeller burnout.

6. Accessing the Cloud Dashboard

The Drop2Crop web application is hosted live via GitHub Pages at: **https://fahimashabb.github.io/drop2crop/**. It connects directly to the MQTT broker over port 8081 Secure WebSockets. Features include real-time SVG animated water level gauges, soil moisture dials, active 24h rain countdowns, interactive timer management (add, toggle, remove slots), and manual pump override controls.

7. Device Online / Offline Status & Hardware Verification

Real-time Online vs. Turned-Off Detection:
The web dashboard tracks hardware status using a dual-layer heartbeat and MQTT Last Will & Testament (LWT) system. The ESP32 transmits telemetry packets every 2 seconds. If you unplug the 5V mobile charger or power is lost, the cloud immediately detects the disconnection within 6 seconds. The status badge automatically updates to a prominent red **'The Device is Not Online (Turned Off)'** warning, and an alert banner shows the exact time elapsed since the controller was last active. In offline state, irrigation commands are safely locked to prevent confusion.

How to Verify That Your ESP32 is Working Properly:

- **1. Red Power LED Check:** When the 5V charger is connected, the solid red **PWR** LED on the ESP32 board must illuminate immediately. If this LED is off, check your wall socket, USB-C cable, or charger adapter.
- **2. Wi-Fi 'Proof of Life' Beacon:** If the device cannot connect to your Wi-Fi within 12 seconds, it automatically broadcasts an open Wi-Fi network named **'Drop2Crop_WiFi_Setup'**. If you see this network in your smartphone's Wi-Fi list, **the ESP32 microcontroller, CPU, and Wi-Fi chip are 100% healthy and operational**.
- **3. Live Web Dashboard Telemetry:** Once connected to your Wi-Fi, the web app at https://fahimashabb.github.io/drop2crop/ turns from red to a pulsing green **'Device Online'** badge, and sensor readings (soil moisture & water tank level) stream live every 2 seconds.
- **4. Physical Relay Mechanical Click:** From the dashboard, switch to **Mode 3 (Manual Cloud)** and press 'Start Pump'. You will hear an audible mechanical *'CLICK'* from the blue relay box, and the red LED on the relay board turns on.

8. Station Security, PIN Protection & Access Control

To prevent unauthorized users or strangers from accessing your farm's interface and turning on the motor, Drop2Crop implements dual-layer end-to-end access control:

- **Web Gateway Lockscreen:** Opening the web application presents a full-screen security gate. Unlocking requires entering your 4-digit PIN. The default factory PIN is **2564** (matching the last 4 digits of device MAC).
- **Hardware-Enforced Command Verification:** Every MQTT command packet transmitted over the internet contains an authentication token. Even if an attacker learns your MQTT broker URL and device topic, the ESP32-S3 strictly rejects any command lacking the secret PIN.

• **Updating Your PIN:** In Dashboard Settings (■■), you can change your PIN at any time. Updating the PIN saves it permanently into the ESP32's non-volatile flash memory (NVS) and your browser session.

• **Instant Lock & Brute-Force Lockout:** Tap the ■ **Lock** button in the top navigation bar to immediately secure the terminal. After 5 incorrect attempts, the gateway locks out for 30 seconds.